

SmolVLM

Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University

Sub-1GB inference | Competitive with 300× larger models | Fully open-source

arXiv: 2504.05299

Large VLMs Are Powerful but Impractical for Edge Deployment



The Edge Deployment Gap

State-of-the-art VLMs demand massive resources:

- **Molmo-A1B-7B: 27.7 GB RAM**
- **InternVL2-2B: ~8 GB RAM**
- **Typical mobile device: 4–8 GB total**

Edge and mobile devices cannot host these models, blocking on-device multimodal AI.

27.7 GB

GPU RAM for Molmo-A1B-7B

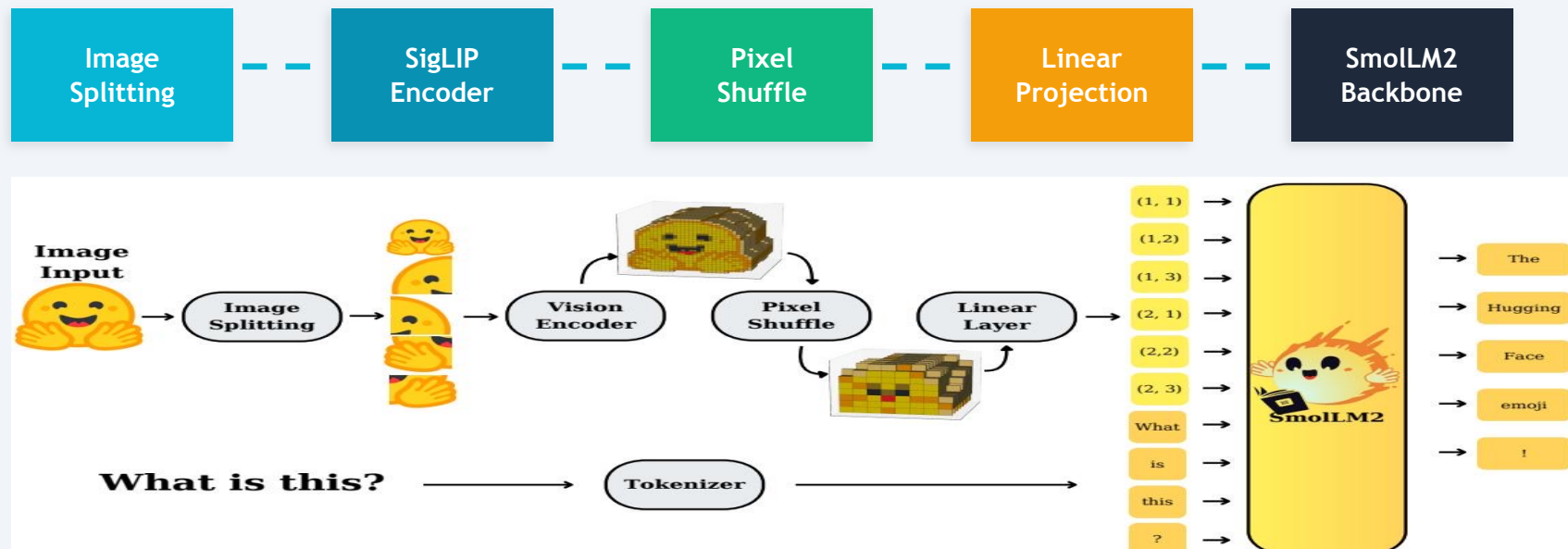
7B+

Parameters in smallest efficient OS model

< 1 GB

SmolVLM-256M RAM usage (batch=1)

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2



1. Images split into sub-images
2. SigLIP encodes visual features
3. Pixel shuffle compresses tokens
4. Linear projection to LM space
5. SmolLM2 generates output

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1



SigLIP-B/16 (93M params)

Paired with 135M LM

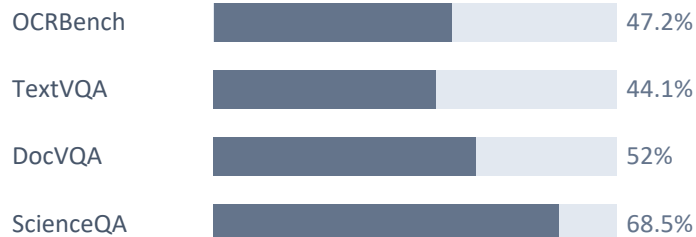


Outperforms



SigLIP-SO400M (428M params)

Paired with 135M LM



4.6× more params

Key insight: Only at 1.7B LM scale does the larger 428M encoder justify its 10% parameter overhead.

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Finding 3

Pixel Shuffle Mechanism

Before (r=1):



256 tokens



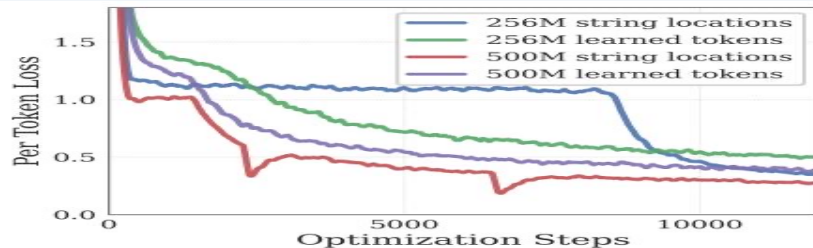
After (r=4):



16 tokens

16×

fewer



How Pixel Shuffle Works

Rearranges spatial features into channels (space-to-depth), reducing visual tokens by r^2 .

- **Small VLMs: $r = 4$**
 - 16× token reduction, eases attention overhead and improves long-context modeling
- **Larger VLMs: $r = 2$**
 - 4× token reduction preserves more spatial detail for higher-capacity models

Why it matters:

Fewer tokens → lower attention cost → better long-context modeling → higher resolution images fit in limited context window

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Finding 2

2K tokens



Baseline
SmolLM2 default

8K tokens



SmolVLM-256M
& 500M

16K tokens



SmolVLM-2.2B
Best performance



RoPE Base Adjustment

Increased RoPE base from 10K to 273K to support positional encoding at longer sequences.

Fine-tuned on mixed long/short-context data to maintain performance on shorter sequences.



Impact on Performance

Significant performance improvement from 2K → 16K context:

- Enables higher-resolution image inputs
- Supports multi-frame video understanding
- **2.2B model uses 16K; 256M/500M use 8K**

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Capability	Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS Baseline
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo)
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo)
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo)
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo)
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo)
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo)
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo)
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo)
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo)
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2)
Video	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2)
Average	All Benchmarks	44.0%	51.0%	59.8%	—

0.8 GB

256M RAM

1.2 GB

500M RAM

4.9 GB

2.2B RAM

27.7 GB

Molmo-7B RAM

SmolVLM-256M surpasses the 300× larger Idefics-80B

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon



HuggingSnap iOS App

Entirely local inference — no cloud dependency

~80 tok/s

MacBook Pro M1 Max 64GB

batch=1, 256M model

~55 tok/s

MacBook Air M3 16GB

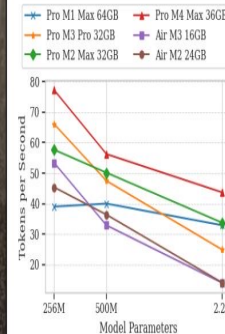
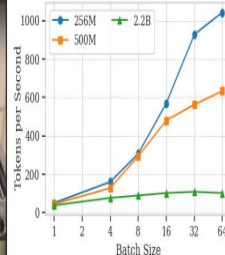
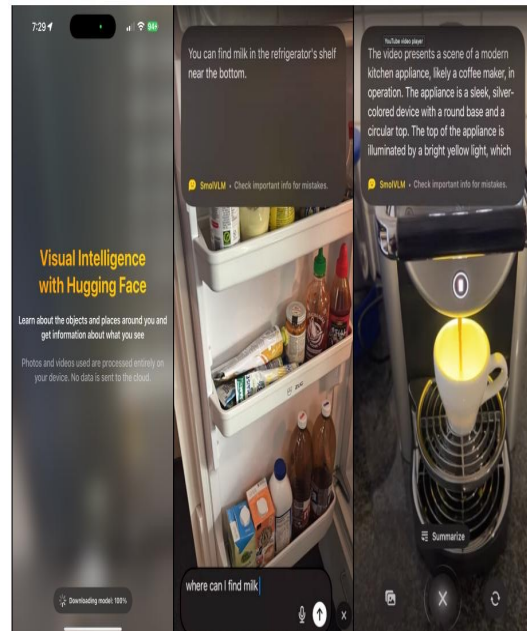
batch=1, 256M model

~1000 tok/s

M1 Max 64GB (batch=64)

256M model, high throughput

Throughput scales roughly linearly with batch size for the 256M model.



Key Takeaways

Design Principles for Efficient Multimodal Models

1

Balance Encoder-LM Compute

Smaller vision encoders (93M) outperform larger ones (428M) when paired with compact LMs.

2

Extend Context Length

Scaling from 2K to 16K tokens via RoPE adjustment unlocks higher-resolution inputs.

3

Aggressive Token Compression

Pixel shuffle at $r=4$ yields 16× reduction for small models, easing attention costs.

4

Sub-1GB Inference Is Achievable

SmolVLM-256M runs at 0.8 GB RAM, surpassing the 300× larger LLaVA-80B.

5

Fully Open-Source Release

All weights, data, code, and mobile app (HuggingSnap) are open-sourced.

Efficient architecture > raw scale • On-device multimodal AI is here